29. Discuss the measurement issues in assessing learning achievement and the problems related to evaluation system. (A) [Ans: 3:13:4 + 3:13:4:01 to the end of 3:13:4:05 + 3:13:5 + 3:13:5:01 to the end of 3:13:5:03]

30. Write a note on 'Reforms in Assessment'. (B) [Ans: 3:14]

31. What is meant by openbook system of examination? Mention its advantages and disadvantages. (B) [Ans: 3:15 + 3:15:1 + 3:15:5 + 3:15:6]

32. Explaining the concept of open book examination system, state its advantages and limitations. (A) [Ans: 3:15 + 3:15:1 to the end of 3:15:3 + 3:15:5 + 3:15:6]

33. State the various misconceptions about the openbook examination system. (B) [Ans: 3:15:4]

34. What is 'online examination'? Explain how it is conducted and mention its advantages and limitations. (A) [Ans: 3:16 + 3:16:1 + 3:16:2]

UNIT – IV

ASSESSMENT PRACTICES IN INCLUSIVE SCHOOL

4:00 Introduction

'Inclusive Assessment' means an approach to assessment of pupils' learning in a mainstream setting, wherein the policies and practices followed facilitate all kinds of pupils, both normal and disabled, learn together as far as possible and improve their respective learning. **'Inclusive Assessment Practices'** involve adopting policies and practices that facilitate employing differential assessment methods and techniques to suit normal, gifted, slow learners and disabled pupils who learn together in the class as far as possible and enable each one of them to improve their learning.

In this unit, concept of differentiated assessment and its definition, principles of differentiated assessment, concept of culturally responsive assessment, ways and means of organizing culturally responsive assessment, use of tests for learner appraisal, construction of an achievement test, construction of a diagnostic test, scoring key and marking scheme, qualities of a good test, ensuring fairness in assessment, assessment for enhancing confidence in learning, assessing the disabled and performance outcomes of diverse learners, Assessment and feedback, Process of feedback and feedback strategies, are to be discussed in detail.

4:01 Differentiated Assessment

Individual students differ in their rate of learning and style of learning. Hence, psychologists advocate that individual differences are to be taken care of by arranging classroom instruction to suit the needs and abilities of each student in the class. **Just as teaching, in the assessment of students' learning too, if individual differences are taken care of, then such assessment practice is known as 'Differentiated assessment'.** Teachers use differentiated assessment to match and respond to the varying learning needs of diverse students in a classroom.

4:01:1 Meaning and Definition of Differentiated Assessment

Differentiated assessment is defined as the way by which teachers make adjustments to and modify assessment activities for individual students or a group of students to cater for different learning needs and a range of learning styles and preferences.

Differentiated assessment may take into account the differences between individual students, such as their

- current level of ability to understand a topic or skill

- prior learning experiences
- learning styles and preferences
- motivation and engagement with learning
- interest and talents

4:01:2 Principles of Differentiated Assessment

Differentiated assessment can lead to enhanced student-learning as they use their current understanding to discover, construct and incorporate new knowledge and skills. It involves teachers considering a range of assessment opportunities to suit the needs, interests and abilities of individual students.

Differentiated assessment involves the following :

i) teacher collecting data regarding individual student's prior knowledge, learning ability, learning needs and preferences etc. before teaching a topic or content unit.

ii) teacher undertaking differentiated assessment based on the data collected about individual students i.e. the teacher divides the students into many groups based on their learning needs and preferences, learning ability etc. and arranges for a suitable instructional method for each and every group of students. Individualized instruction will be arranged for students who greatly differ from other students in their learning ability.

iii) Considering what resources and stimulus materials will be highly useful to individual groups and making use of them accordingly, while teaching is undertaken.

iv) Undertaking formative assessment which consists of many techniques such as asking questions intermittently during teaching, providing a short assignment daily, based on what are taught in the class that day, using oral tests during and at the terminal stage of teaching etc.

v) Providing individualized feedback to students, based on the findings obtained in the summative evaluation of learning attainment, held after the instructional period so as to make the students understand their respective strengths and weaknesses, in the content areas taken for assessment.

Putting briefly, differentiated assessment involves (i) the teacher collecting data regarding students' learning, before commencing his teaching, (ii) then based on the collected data, employ suitable instructional methods and techniques that cater to the learning needs of various groups of students in the class, (iii) using different assessment tools and techniques like teacher made written tests, standardized tests, performance based tests, checklists of traits and characteristics that suit

individual needs of students, to evaluate the learning achievement of each student in the class, after the instructional period and (iv) providing individualized feedback to students to improve their learning achievement.

4:02 Culturally Responsive Assessment

Schools in India today are becoming increasingly diverse, consisting of students coming from different cultural backgrounds i.e. students in our classrooms differ with respect to their race, religion, language, socio-economic status etc. For example, in Tamilnadu schools, apart from tamil students we can find students having Malayalam, Telugu or Kannada as their mother tongue. Similarly in our schools we can find, students hailing from Anglo-Indian families as well as those with French cultural background. We can also spot Negro students, hailing from some African countries in our city colleges. Research studies point out that cultural background of students has significant influence on their learning. A good learning environment is one where students who are culturally and linguistically diverse are provided with equal opportunity to learn and grow. Hence, the assessment practices to evaluate learning achievement of culturally different groups of students should also be flexible enough to cater to their needs. An assessment practice which functions in such a way that none of the cultural groups among the students in the class feel affected in any way is called **'Culturally Responsive Assessment'.**

4:02:1 Ways and Means of Creating Culturally Responsive Assessment

Social economic status, native language and learning styles of students should be taken into account in creating culturally responsive assessments.

1. Socio-economic Status

Research findings reveal that students from families of high socio-economic status in which one or both parents have a college education tend to have more advanced linguistic capabilities and accomplishing school tasks comes more easily than students from economically disadvantaged homes. Hence it is necessary to provide special training to students hailing from low socio-economic strata of the society, in facing examinations.

2. Native Language

If the language spoken at home and the one in the place of living (Native language) are one and the same, then no complexity arises in educating the child through the native language. Instead, if the child is educated through a language other than its mother tongue i.e. native language or any foreign language, then the child finds it difficult to learn that language particularly in pronouncing the words properly as well

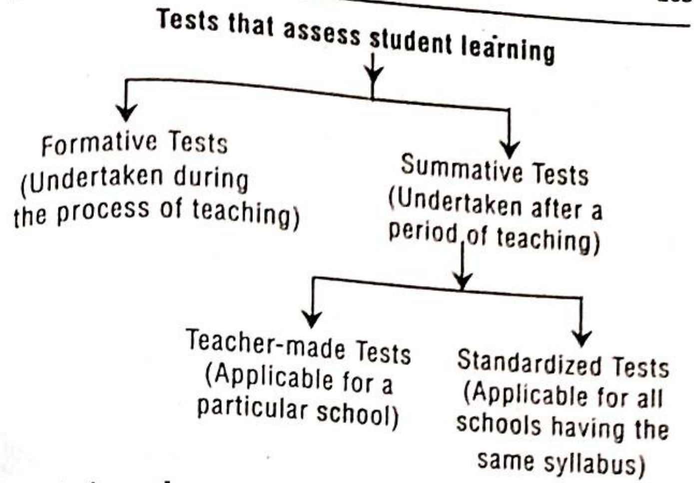
as writing and understanding its traditional language style. Hence, as far as possible, arrangements should be made for children to face examinations in their own mother tongue; otherwise while valuing the answer scripts, marks should be awarded only for pupils' understanding of the subject content and no importance should be given for the style of language expressed and the spelling of words used in their answers.

3. Learning Style

The learning style of each student differs. Some prefer to learn through visuals; some others find it easy to learn by doing; few others may have the ability to learn through abstract thinking i.e. students differ in their learning styles. If the assessment of students' learning achievement tries to take into account the learning style of each student, then it is termed as 'Culturally responsive' assessment. For example, a student performing well in written tests, may not be doing well in oral tests or performance tests that involve physical skills. Therefore, a common uniform method of assessment for all is not proper; it is only proper to have culturally responsive assessments to evaluate students' learning achievement by employing different methods and techniques and that too in the language in which students feel they are more comfortable. It may appear to be impossible at present to tailor assessment techniques to suit the needs of individual students, but as technology improves, the tools may come into existence to make this atleast partially a reality.

4:03 Use of Tests for Learner Appraisal

Tests serve as the touch stones for the appraisal of learners' achievement and its quality. Assessment and teaching - learning process are integrated with each other. Tests that are directly related to the teaching - learning process are of three types. They are : (i) Formative assessment tests (ii) Achievement tests and (iii) Diagnostic tests. Tests that evaluate students' learning as it develops during the period of teaching, are known as 'formative tests'. Tests that help to assess the learning achievement of students, after the teaching is over, are called 'summative test'. Summa -tive tests are also known as 'achievement tests' or 'Examinations'. Based on the findings of achievement tests, diagnostic tests are conducted to identify the specific nature of difficulties experienced by students in certain areas of the subject content taught and arrange for 'remedial teaching' accordingly. Achievement tests are of two types - (i) Teacher made tests applicable for a particular school and (ii) Standardized tests applicable for all schools which follow a particular syllabus; their validity and reliability are well established and often they will be at a high level.



4:03:1 Important Features of Using Tests for Learner Appraisal

i) The process of testing the quality of students' learning is based on evaluating the responses offered by students for the various test items which reflect their understanding of the subject content.

ii) Appraisal of students' learning is done by matching their learning achievement with the expected learning outcomes.

iii) Rate of students' learning reveals how far the classroom instruction of the teacher has successfully met the learning needs of students.

iv) Appraisal of students' learning is undertaken for improving their learning.

v) It is the duty and responsibility of the teacher to conduct periodically student appraisal with regard to their learning and report the same to their parents and other stake holders in education.

vi) Teacher has to employ a variety of assessment tools and techniques to assess student learning.

vii) Assessment of student learning is not an annual event but integrated with the process of teaching.

viii) Learner appraisal is a process that is to be undertaken carefully and efficiently.

4:04 Construction of an Achievement Test

Constructing a test to assess how far students have achieved proficiency in learning the course content in a subject i.e. constructing an achievement test in a subject of study, occupies an important place in educational evaluation. The following are the major steps involved in the construction of an achievement test.

1. Planning the test
2. Determining the design of the test
3. Preparing the 'Blue-print' of the test
4. Writing the test items and preparing the question paper

Let us discuss the above steps in the construction of an achievement test in detail in the following sections.

4:04:1 Planning the Test

The important tasks involved in this step are :

a) Deciding the content area for the test

b) Deciding the time-limit for answering the test

c) Fixing the total marks allotted for the test

d) Deciding the types of test items (i.e. Essay type questions, Paragraph questions, Short-answer questions and Objective type questions) to be included in the test.

e) Deciding about the number of items in each type of question.

f) Deciding about providing choices for students in answering questions.

4:04:2 Determining the Design of the Test

Tasks involved in this step are :

i) Identification of instructional objectives to be tested and assigning weightage to each of them.

ii) Assigning weightages to the content areas to be tested.

iii) Assigning weightages to different forms of test-items.

iv) Assigning weightages to difficulty levels.

4:04:2:01 Assigning Weightages to Instructional Objectives

Generally, in achievement tests instructional objectives like 'Knowledge', 'Understanding', 'Application' and 'Skills' will be tested. Determining the amount of importance to be given to these objectives in terms of percentage of marks to be allotted is known as **'Assigning weightages to instructional objectives'**. In a good achievement test, for the two instructional objectives 'Knowledge' (consisting of the components 'recall' and 'recognition') and 'Understanding' put together, not more than 60% of the total marks of the test will be allotted; for the objective 'application' 30% of marks and the remaining 10% of marks for 'skills' will be allotted.

(E.g.) In the model 'Blue-print' given, weightages assigned to the various instructional objectives are as follows :

S. No.	Instructional Objectives	Marks Allotted	% of Marks Assigned
1.	Knowledge	10	20
2.	Understanding	20	40
3.	Application	15	30
4.	Skill	5	10
	Total	50	100

4:04:2:02 Assigning Weightages to Content Areas Tested

Allotting percentage of marks to the different topics or lessons in the content areas (or content units) taken up for the test, depending upon their importance is otherwise known as **'Giving Weightages to Contents'**.

(E.g.) In the model 'Blue-print' given, weightages assigned to the six content units taken up for the test are as follows :

S. No.	Content Units	Marks Allotted	% of Marks Assigned
1.	Unit I	5	10
2.	Unit II	6	12
3.	Unit III	18	36
4.	Unit IV	5	10
5.	Unit V	10	20
6.	Unit VI	6	12
	Total	50	100

4:04:2:03 Assigning Weightages to Different Forms of Test-Items

The number of test-items to be included in a test depend upon the (i) Time limit fixed for the test (ii) Forms of test-items to be included (iii) Students' age level (iv) Difficulty level of the contents and (v) The range of the subject content. Upon deciding on the number of test-items to be included in the test, the next step is to decide about the different forms of test-items that should find place in the test.

For a test of one hour duration, three or four types of questions and for a two hour test, five or six forms of test items could be included. For a test of one period of about 45 minutes, it is better to use only three types of questions – (i) Essay type question (ii) Very short answer type questions and (iii) Objective type test-items. Among the objective type test-items, it is better to use the multiple choice type; if possible among the types, 'Fill-up the blanks', 'True or False', 'Matching' etc. any one or two forms may be included in the test. It is important to keep in mind that within the allowed time, students should be able to answer all the questions.

The common practice followed is to assign one mark for each objective type test item, two marks for very short answer type item, four or five marks for a paragraph question (i.e. short answer type) and 8 or 10 marks for an essay type question.

(E.g.) In the model 'Blue-print' given for a test of one hour duration, weightages given to different forms of test-items are as follows :

S. No.	Form of Test-Item	No. of Test-Items	Allotted Marks	% of Weightage given	Time required to answer (in minutes)
1.	Essay Type Question	1	10	20	20
2.	Paragraph Question	2	10	20	10
3.	Very Short Questions	5	5	10	5
4.	Objective Type Question				
	a) Multiple Choice	10	10	20	10
	b) Matching	5 (Premises)	5	10	5
	c) True/False	5	5	10	5
	d) Fill up the blank	5	5	10	5
	Total	**33**	**50**	**100**	**60**

4:04:2:04 Assigning Weightages to Difficulty Levels

Generally, most of the items in an achievement test (covering about 60% of total marks) will be of 'average difficulty level'; those covering 20% of total marks will be difficult to answer, serving as a challenge for the gifted and the rest of the test-items (i.e. those covering 20% of marks) will be 'easy' to answer for all.

(E.g.) In the model 'Blue-print' given, weightages assigned to the different difficulty levels of test-items are as follows :

S. No.	Difficulty level of Test-items	Marks Allotted	% of Marks Assigned
1.	Difficult Questions	10	20
2.	Questions with 'average difficulty level'	30	60
3.	Easy Questions	10	20
	Total	50	100

4:04:3 Preparing the 'Blue-Print' of the Question-Paper

'Blue-print' is a three dimensional chart showing (i) instructional objectives tested (ii) content areas included and (iii) type of test-items finding place in the test along with number of test items and marks assigned for each of the three dimensions. The 'blue-print' reveals the structure of the question paper that is to be prepared.

Model Blue-Print

Instructional Objectives → / Content Units ↓	Knowledge				Understanding				Application				Skills				Total
	E	S.A.	V.S.A	O	E	S.A.	V.S.A	O	E	S.A.	V.S.A	O	E	S.A.	V.S.A	O	
Unit – I						5(1)											5(1)
Unit – II				1(1) (T/F)				5(5) ((Mat.)									6(6)
Unit – III				2(2) (F.B)	7(1)		1(1)	1(1) (M.C.)			1(1)	3(3) (M.C.)	3(1)*				18(9)
Unit – IV										3(1)				2(1)*			5(1)
Unit – V				2(2) (T/F)							2(2)	6(6) (M.C.)					10(10)
Unit – VI				5(5) (F.B-3 T/F-2)			1(1)										6(6)
Total	–	–	–	10(10)	7(1)	5(1)	2(2)	6(6)	–	3(1)	3(3)	9(9)	3(1)*	2(1)*	–	–	50(33)
	10(10)				20(10)				15(13)				5(–)				

Note : 1. E denotes Essay Type; S.A. → Short Answer Type (Paragraph); V.S.A. → Very Short Answer; O → Objective Type

2. Mat → Matching; M.C. → Multiple Choice; T/F → True/False; F.B. → Fill up the Blank

3. Number inside the Braket → Number of test items; *Number outside the bracket → Marks allotted*

4. * denotes part of a question

4:04:3:01 Use of Blue-Print of the Question-Paper

i) Use of blue-print in the preparation of question paper increases its content validity.

ii) It clearly indicates the scope of the test and the extent of importance given for the various aspects of it.

iii) It ensures that due importance is given for the learning outcomes and subject contents.

iv) It ensures that each test-item, tests a learning outcome.

v) It ensures that each test item is independent and free from overlapping.

4:04:4 Writing the Test-items and Assembling the Question Paper

After writing the test-items from different topics or units of the subject content as per the blue-print, each type of questions are to be grouped separately while assembling the test-items and giving a final shape to the question paper. Appropriate instructions regarding how to answer the questions in a given group should be stated briefly.

[Note : Difficulty level of a test-item is to be found by resorting to item-analysis. This method is used in the preparation of a standardized achievement test. But in the construction of a teacher-made achievement test, difficulty of test items are determined purely based on his/her professional experience. Further, if majority of the testees could answer an item correctly, then it is taken to be an easy item. If a test-item could be answered only by a few students, then it is presumed, it is a difficult item].

4:05 Scoring Key and Marking Scheme

After the preparation of a question paper, the immediate task to follow is developing the scoring key and the marking scheme that provide the guidelines for awarding marks for students' answers for each of the test-items in the question paper.

4:05:1 Scoring Key

The list containing the correct answers for the objective type test-items of the question-paper is known as 'Scoring-Key'. Students' answers for the test-items are to be compared with the correct answers given in the scoring key in awarding marks for their responses accordingly.

Example for a Scoring Key

Question No.	Correct Answer	Marks to be awarded
1.	(a)	1
2.	(c)	1
3.	(d)	1
4.	(b)	1
5.	True	1
6.	False	1
7.	False	1

4:05:2 Marking Scheme

'Marking Scheme' is to be prepared and provided to the scorers of the answer scripts, indicating the guidelines for awarding marks to questions like 'short-answer type', 'answer in about a paragraph', 'essay-type' etc. i.e. it indicates the points or steps expected in the answer for each question and the distribution of the total marks of that question for the expected points / steps. In addition to these, for language test papers the marking scheme will indicate how many marks are to be deducted for spelling mistakes and grammatical errors; for science test-papers, it will be indicated how many marks are to be deducted for not expressing the measurements in the proper units like 'kilogram', 'metres', 'seconds', 'watts', 'volts' etc.

Example for a Marking Scheme

Test No.	Type of the test-item	Expected Answer	Marking Scheme
8.	Very short -answer type	Two reasons that are expected i) ______________ ii) ______________	One mark is to be given for each correct reason.
9.	Short -answer type	Answers which fall in a paragraph or half a page	
		i) Full marks can be given if all the expected points (four or five) are provided correctly	Full marks of 5 could be awarded.
		ii) Out of the 5 points, if three are stated correctly	Only 3 marks to be awarded.
		iii) If, only one point is correctly stated in the answer	1 mark is to be awarded

Test No.	Type of the test-item	Expected Answer	Marking Scheme
10.	Essay	Expected answer will be in two or three pages. In it, appropriate points will be indicated as (i), (ii), (iii) etc.	
		i) If all the points are correct	Full marks 10 can be given.
		ii) Expected answer has eight points out of which only, 4 to 6 points stated in the student's answer are correct.	5 to 7 marks can be given.
		iii) If less than 4 points stated in the student's answer are correct.	3 marks can be given

Evaluation of answer scripts using 'Scoring Key' and 'Marking Scheme' ensures to a large extent, objectivity and transparency in the scoring.

4:06 Construction of a Diagnostic Test

Before the construction and use of a diagnostic test by a teacher, students who struggle to learn the subject content are to be identified. Generally by analyzing the achievement test scores, the teacher could identify easily the students who lay behind significantly in their learning achievement.

After arranging the achievement test scores of students in the descending order, bottom 10% of students could be identified as backward in learning.

It could be easily inferred that they are having learning difficulties.

Following are the five steps in the construction of a diagnostic test that helps to identify the students with learning difficulties, the content areas which they find it difficult to learn and the nature of their learning difficulties.

1. Planning
2. Writing the test items
3. Assembling the test items to form the question paper
4. Providing clear instructions for answering the test
5. Providing the scoring key and marking scheme.

4:06:1 Explanations for the Different Steps Involved in the Construction of a Diagnostic Test

The five steps involved in the construction of a diagnostic test are explained below :

4:06:1:01 Planning the Diagnostic Test

By undertaking questionwise analysis of students' marks in an achievement test, the teacher can easily identify the questions for which more than 50% of students in the class have given wrong answers and the content areas in which these questions find place, are taken to be the areas of learning difficulties of students. The teacher then should make exhaustive content analysis, in the areas of learning difficulties.

Each content should be divided into small learning points. No learning point should be ignored as insignificant. The learning points thus found out are to be arranged sequentially in a logical order. Atleast one test-item is to be developed for testing each learning point.

For example, in the physics lesson on 'Thermometers', if it is found that students find it difficult to convert the given measurement in 'Centigrade Scale' to 'Fahrenheit Scale', then this content should be broken into small learning points as follows :

i) Lower Fixed Point
ii) Upper Fixed Point
iii) No. of divisions between the upper and lower fixed points in both the thermometric scales.
iv) The reading of origin in the two kinds of thermometric scales
v) Formula for converting °C into °F [F° = (9/5 x C°) + 32]

4:06:1:02 Writing the Test Items of the Diagnostic Test

Generally, the test-items of a diagnostic test are in the form of 'Very short answer type' or 'Objective type' questions. Questions should be stated in short sentences with simple words, bringing out the learning points clearly. Questions should have only one correct answer. For bringing out each learning point, suitable questions are to be framed. Questions should take the learner gradually from the known to the unknown learning points.

4:06:1:03 Assembling the Test-items to form the Question-Paper

Questions are to be assembled in a logical sequence and the number of each question should be mentioned clearly.

4:06:1:04 Providing Clear Instructions for Answering the Test

Clear-cut directions for answering the test should be mentioned briefly and precisely. Generally, no time limit is fixed for answering the diagnostic test in full. It should be emphasized that testees should answer all the test-items in the question-paper. It should be mentioned clearly how to mark the answers and that no detailed answer is necessary for any of the questions; or instructions such as "you can mark your answer in the question paper itself" may be provided. Students are required to write their name on the question paper.

4:06:2 Model Diagnostic Test Question-Paper

S. No.	Questions	Marks
1.	Draw the diagram of a Centigrade thermometer marking important measurements in it.	5
2.	What is the lower fixed point in the Centigrade scale?	2
3.	What is the upper fixed point in the Centigrade scale?	2

S. No.	Questions	Marks
4.	How many divisions are there in between the lower and upper fixed points in the Centigrade scale?	1
5.	Draw the Fahrenheit thermometer, marking the important measurements in it.	5
6.	What is the lower fixed point in the Fahrenheit scale?	2
7.	What is the upper fixed point in the Fahrenheit scale?	2
8.	How many divisions are there in between the lower and upper fixed points in the Fahrenheit scale?	1
9.	100 divisions of Centigrade scale are equal to how many divisions of the Fahrenheit Scale?	1
10.	1 division in the Centigrade scale is equal to how many divisions of the Fahrenheit Scale?	1
11.	5°C equals to how many divisions of the Fahrenheit Scale?	1
12.	What is the starting point in the Fahrenheit Scale?	1
13.	5°C will be equal to what measurement in the Fahrenheit Scale?	1
14.	5°C = ... °F	1
15.	What is the general formula for converting Centigrade measurements into Fahrenheit measurements?	2
16.	20°C = ... °F	2

4:06:3 'Scoring Key' and 'Marking Scheme' for the Given Model Diagnostic Test

Q. No.	Scoring Points / Cues	Methods of awarding Marks
1.	i) Drawing the diagram of the Centigrade Thermometer	Bulb part-1; body part-1
	ii) Labelling the lower fixed point	1
	iii) Labelling the upper fixed point	1
	iv) Pointing out the readings lying in between the two fixed points	1
2.	0 °C	Numerical Value - 1 Mentioning the unit 0°C-1
3.	100 °C	Numerical Value-1 Mentioning the proper unit-1
4.	100	1
5.	i) Drawing the diagram of the Fahrenheit Thermometer	Bulb portion-1; body part-1
	ii) Labelling the lower fixed point	1
	iii) Labelling the upper fixed point	1
	iv) Pointing out the readings lying in between the two fixed points	1
6.	32 °F	Numerical Value-1 Mentioning the unit °F-1
7.	212 °F	Numerical Value-1 Mentioning the proper unit -1

Q. No.	Scoring Points / Cues	Methods of awarding Marks
8.	180	1
9.	100 divisions of Centigrade scale = 180 divisions of Fahrenheit scale	1
10.	$\frac{180}{100} = \frac{9}{5}$	1
11.	$5 \times \frac{9}{5} = 9\ °F$	1
12.	32 °F	1
13.	9+32 = 41st division in the Fahrenheit Scale	1
14.	41	1
15.	F = (C x 9/5) + 32	Writing C x 9/5 –1 Writing (Cx9/5) + 32 –1
16.	F = (20x9/5) + 32 = 36+32 = 68°F	Writing the value as 68 – 1 Mentioning the Proper Unit - 1

4:06:4 Diagnosing the Nature of Learning Difficulties and Taking up Remedial Measures

4:06:4:01 Diagnosing the Nature of Learning Difficulties of Students

After administering the diagnostic test and scoring the answer scripts using the marking scheme, the

'diagnostic table' should be prepared. In preparing the table, if a student gets half or more of the total marks for a question, then it should be recorded with the symbol (✓); and if gets, less than half of the total marks for the question, then the symbol 'X' to be put.

From the scores obtained for the given model diagnostic test, the diagnostic table is prepared as follows :

Student's Name	Question Number															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	x	x
B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	x	x	x
C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	x	x
D	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	x	x
E	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	x	x
F	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	x	✓	x	x
Total	6	6	6	6	6	6	6	6	6	6	6	6	5	5	0	0

In the above table, we can find question number 15 and 16 have been answered incorrectly by all the students. On examining further, we can understand that students do not follow the BODMAS Law (Bracket, of, Division, Multiplication, Addition, Subtraction) correctly.

4:06:5 Taking-up Remedial Measures

Remedial measures are all not, of the same kind; they differ as per the nature and causes of learning difficulties experienced by students. Some kind of learning difficulties could be remedied by taking measures like 'Revision' and 'Reteaching'.

'Reteaching' does not mean 'teaching again'; it means the teacher enriching what he taught lacing with further explanations, additional illustrations, undertaking demonstrations, providing individual attention to students, increasing the student participation in the teaching-learning process, and providing more amount of drill and practice for students.

Teacher can increase students' interest in learning through measures like adopting interesting approaches in classroom teaching, providing individual attention to students, monitoring students' learning etc. Generally remedial measures improve student learning; some students who are highly backward in their learning (slow learners), may need individualized instruction. Students having the same kind of learning difficulties could be grouped together and for them remedial measures like providing oral practices, writing work and computer assisted instruction, participation in certain co-curricular activities etc. could be undertaken.

In our example, the teacher can explain in detail how to use the BODMAS rule with illustrations and give adequate amount of drill and practice so that the students identified with learning difficulties in the conversion of Centigrade measures into the Fahrenheit scale and vice-versa could achieve enough learning proficiency.

4:07 Questionwise - Analysis / Item - Analysis

The quality of a test depends upon the test items contained in it. Therefore test-items are to be carefully selected, particularly while developing a standardized test, by finding the discriminative index and difficulty level of each test-item that is considered for selection, to be included in the final form of the test. Thus 'Item-analysis' is concerned with the problem of identifying the best items that are to be included in the final form of the test so that the test will have certain specified characteristics like validity and reliability'.

4:07:1 Indices of Difficulty Level and Discriminative Index of Test-Items

After scoring the answer scripts of the test in the pilot study, based on the total marks, individuals are placed in order, from high to low (i.e. descending order). Top 27% constitutes the 'high achieving group' and the bottom 27% constitutes the 'low achieving group'.

We have then to find for each test-item how many students in each of the two types of groups have answered correctly and compute the difficulty level and discriminative index of the test items.

$$\text{Difficulty level of the test item (DL)} = \frac{\text{Total number of students in both the groups, who have answered the test item correctly}}{\text{Total number of students in both the groups put together}}$$

Taking the difficulty level value .50 as the dividing line, those items having value less than .50 are taken to be 'difficult items' and those having value more than .50 are considered as 'easy items'.

Discriminative index (D.I) of a test-item refers to the ability of the test-item to discriminate a high achiever from a low achiever. For a test-item having high discriminating power, number of students answering it correctly will be more in number in the high achieving group and less in low achieving group. Discriminative index (D.I.) for a test-item is calculated using the following formula.

$$DI = \frac{\text{No. of students who answered the test-item correctly in the high achieving group} - \text{No. of students who answered the test-item correctly in the low achieving group}}{\text{No. of students in the high or low achieving group}}$$

(E.g.) Suppose among 50 students who participated in a pilot study, 9 of the high achieving group and 5 of the low achieving group have answered correctly for a particular test-item. In that case,

No. of students in the High-Achieving group = $\frac{27}{100}$ x50=13.5 $\simeq$14

Similarly No. of students in Low-Achieving group = 14

$$\therefore \text{D.I.} = \frac{9-5}{14} = \frac{4}{14} = .285 \simeq .29$$

$$DL = \frac{9+5}{14+14} = \frac{14}{28} = .50$$

[Note : The above method of calculation is appropriate for objective type questions only. For essay type questions the following method should be used]

$$\text{Difficulty Level (DL)} = \frac{\text{Total marks obtained for a particular test item (like essay type)}}{\text{Maximum Marks for that test-item} \times \text{No. of those who attempted to answer that test-item}}$$

$$\text{D.I.} = \frac{X_1 + X_2 + X_3 + \ldots\ldots}{\text{Maximum Marks for that test-item} \times \text{No. of students who attempted to answer that test-item}}$$

Here X1, X2, X3, ... refer to the marks obtained by different students for a particular test-item.

4:07:2 Preparing the Final Form of the Question Paper

Appropriate test-items are to be selected first, based on the discriminative index of the test-items; then from among them, those items having suitable difficulty level are to be selected for the final form of the test. Generally 50% of the test-items selected for a question paper should be of 'average difficulty level', 25% of the items are 'easy', another 20% of the items are 'difficult' and the remaining 5% are 'very difficult'. Items, thus selected should be arranged in the increasing order of difficulty level. For selecting the test-items that are to be included in a question paper, the following table could be used.

Discriminating Power		Difficulty Level	
0.4 & above	Excellent Item	Between 0.4 & 0.6	Average Difficulty
Between 0.4 & 0.3	Good	Between 0.2 & 0.4	Difficult Item
Between 0.2 & 0.3	Average Item	Between 0.6 & 0.8	Easy Item
Between 0.2 & 0.1	Requires Improvement	Between 0.8 & 1.0	Very Easy Item
Less than 0.1	Item to be dropped or condemned.	Between 0 & 0.2	Very Difficult Item

4:08 Qualities of a Good Test

The qualities mentioned as characteristics of a good test are : (i) Validity, (ii) Reliability (iii) Objectivity (iv) Practicability (v) Comprehensiveness and (vi) Clarity.

4:08:1 Validity

By test validity we mean the degree to which the test actually measures what it claims to measure. It tells us what can be inferred from test scores. It is specific to the purpose and the group for which a test is to be used. For example, an achievement test in science should measure students' achievement in science only; there is no place for testing language skill in that. Similarly a test of general intelligence which has been validated with rural students can not

be used for the purpose of knowing the general intelligence of urban population and vice versa. So before administering the test, we should know 'for whom' and 'for what purpose' the test was developed.

4:08:2 Reliability

The reliability of a test is the consistency of scores obtained by the same person when retested with the same / identical test or with an equivalent form of the test. Reliable tests, whatever they measure, yield comparable scores upon repeated administration. Thus reliability of a test is related to its ability to accurately measure, whatever it claims to measure.

4:08:3 Objectivity

Administration of the test, scoring of responses and interpretation of scores obtained, should be independent of the subjective judgement of the individual examiner. i.e. the scores obtained by an individual should be identical regardless of who happens to be the examiner.

4:08:4 Practicability

This refers to the matter concerning the administration of the test. The practicability of a test is high if it can be easily administered, at a short time, with less expenses and also easy to interpret the obtained test scores.

4:08:5 Comprehensiveness

A test should cover all aspects of what are to be tested. In an achievement test, questions should spread across all the content areas included for the test. The provision for the testees to choose among the test-items and respond should be very less. Test-items should test all the instructional objectivities like knowledge, understanding, application, skills etc.

4:08:6 Clarity

Instructions given should be brief and clear so that students could answer without any confusion. Similarly questions should be stated with simple, clear and unambiguous terms. For example, it is better to state the question 'Draw the cross-sectional view of the human heart and name the parts' instead of 'Draw the cross sectional view of the human heart and label the parts'.

A good test need not necessarily possess all the desirable qualities but it is more important to have atleast the three characteristics validity, reliability and objectivity.

4:09 Ensuring Fairness in Assessment

4:09:1 Meaning of Fairness in Assessment

A 'Cricket Pitch' is said to behave fairly if it does not favour more for the bowlers or batsmen but offer

help equally well for all those who operate on it viz. bowlers, batsmen and fielders. Similarly "a fair assessment is one in which all students are given equitable opportunities to demonstrate what they know" (**Lam**, 1995).

Fair assessment means providing equitable opportunities to all students to demonstrate the extent of their learning. It does not mean treating all students exactly in the same way. Fair assessment means that all students are assessed using the methods and procedures most appropriate to them. This kind of assessment may vary from student to student, depending on their prior knowledge, cultural experience and cognitive style. Therefore the assessment system should consists of different methods of assessing students' learning achievement such as oral test, performance / practical test, observation, students' assignments and learning portfolio, project work submitted etc.

4:09:2 Steps to be Taken to Make Assessment Methods Fair

Creating custom-tailored assessments for each student is, largely impractical, but nevertheless there are seven steps we can take, to make our assessment methods as fair as possible. They are :

1. **Having clearly stated learning outcomes**

Only if expected learning outcomes are clearly stated and made known to the students, they understand what are expected of them to learn and master and accordingly get themselves ready. The teacher should provide students the list of concepts and skills that are going to be covered in the assessment and the rubrics to be used to assess the projects and assignments.

2. **What are taught to students alone should find Place in Assessment**

If the teacher expects his students to demonstrate good writing skills, he should without assuming that they would have already developed those skills, explain what constitute 'good writing', provide for enough practice and then only attempt to evaluate their writing skills.

3. **Using many different kinds of measures and many different methods**

In the present day assessment, common written examinations are conducted for all students, based on which significant decisions like passing a course, certification of the level of proficiency achieved etc. are taken, which is not fair. How can we say with confidence that some one getting an average score

of 60 in the written examination is to be placed in 'First Class' for his learning proficiency and another on getting an average score of 59 is to be placed in 'Second Class'. This is not fair. During the entire course duration, information from a broad variety of assessments involving oral tests, written tests, practical examinations, observation, interview, assignments, projects etc. should be obtained, based on which the final decision regarding the learning proficiency level of each student is to be arrived at. Repeating the same assessment technique many number of times does not mean 'using many different measures' in assessment. It simply means using different kinds of assessment techniques that involve reading, writing, doing, creating, investigating in collaboration with peers etc. and that too under taken through out the course / grade. Stated briefly, undertaking 'Continuous and Comprehensive Evaluation' during the entire learning period is the way to ensure fair assessment of students' learning achievement.

4. **Helping Students learn how to do the work that are assessed**

Students hailing from poor and disadvantaged families may have got trained in learning by memorization or drill and practice. Such students are to be helped by the teacher to learn with comprehension and apply appropriately what they have learned while answering the tests, to learn through the use of computer and to develop the ability to take part in computer assisted assessment. Putting it in a nut shell, whatever may be the kind of assessment the teacher undertakes, atleast some students in the class, require his help in learning the skills needed to succeed.

5. Engaging and Encouraging Students

The teacher should encourage students to repeatedly practice what they have learned and express them properly in the assessment tests. Some students may struggle to express their best without the praise and encouragement from the teacher.

6. Interpreting assessment results appropriately

There are two approaches in interpreting assessment results. They are : (i) Judging the achievement of students by comparing them against that of peers (Norm-based evaluation), (ii) Judging the achievement of students on certain standard (Criterion based evaluation).

For example, declaring those who got more than 60% marks as having passed in I Class, 40% of students among those who got their scores less than 60% as having passed in II Class, the next 30% of students as having passed in III Class and the rest of

students as having 'poor achievement' or 'Fail' in the examination, belongs to the first type of approach mentioned above (i.e. Norm-based approach). This is highly inappropriate method of interpreting students' achievement.

In the second type of approach (i.e. Criterion-based approach), level of proficiency of students' learning will be judged on the basis of certain criteria. Declaring those students who complete 75% of the elements constituting a given task correctly as having been placed in the First Division, those who could complete only 60% correctly as having been placed in the Second Division etc. belongs to the second type of approach in interpreting students' achievement. By employing the criterion-based approach in interpreting test scores, the teacher could make the students understand their shortcomings in their learning and give appropriate feedback accordingly to improve their performance.

7. Teacher evaluating his own assessment method

If students do not perform well on particular assessment, the teacher can interact with students, particularly those who scored very low marks, to find out the reasons related to him if any, such as lack of clarity in test-items, inadequacy of instructions provided, or concepts being not taught well in the class etc. The teacher by rectifying these, could increase fairness in his future assessments.

[Note : Most of the books that have come out in the market (both in Tamil as well English) with the title 'Assessment for learning' mention fairness in assessment as unbiased assessment, accuracy in assessment, consistency in assessment, highly reliable assessment, cultural sensitivity in assessment etc. These are the qualities of a good assessment tool and not fairness in assessment].

4:10 Assessment for Enhancing Confidence in Learning

Persisting with efforts to successfully complete a challenging task or giving it up easily, depends on one's self-belief and confidence. Individuals who have high level of self-confidence believe that they could achieve anything, always strive to succeed in what all they do; they can learn almost anything. Regardless of how good a teacher's instruction is, learning of those who function without much self-confidence will never be good or improve. Therefore, the foremost job of a teacher is to develop and maintain confidence in his learners, for which formative assessment helps a lot. The primary purpose of formative assessment undertaken, when the teaching-learning process goes

on in every unit of the subject content, is to improve student-learning.

Good assessment helps the teacher to know about the outcomes of his teaching as well as the students to know about the status of their learning i.e. assessment for learning helps students to know what they have learned and also know how they have been learnt. A student coming to know through formative assessment that his learning progress is inadequate, besides becoming aware of what have been left out in his learning, could also know about the defects in the way he learns. This enables students to develop the ability to control their learning.

Following the formative assessment, students could discuss with their teachers, peers and parents and find the ways and means of improving their learning. This enables students to acquire new knowledge and skills according to their goals and needs.

Building Students' Assessment capability

Teacher should help students understand the following, which enable them to develop the ability to assess their own learning.

i) What high quality work look like?

ii) What criteria define, quality work?

iii) How to compare and evaluate their own work against the criteria of good work?

Teachers while planning their teaching, should keep in mind what kind of learning experiences are to be provided to students and how to assess their learning achievement.

It should be made known to students well in advance, what are the things they are going to learn as well as why and how things learned will be assessed.

Teacher should know by employing a particular assessment method of learning, how well student - learning will get improved. In achieving the instructional and learning goals, assessment for learning helps in the following three ways.

i) Identifying the learning needs

Information obtained from assessment help both the teacher and students. Assessment information help students know their present position of where they are now with respect to their learning, where they want to be and how to reduce the gap between these two positions i.e. what further measures to be undertaken to improve upon their present learning. Similarly assessment infor -mation help the teachers to know what further steps they should take to enable the students to move towards

their goals. Stated briefly, assessment information help to identify the learning needs of students.

ii) Feedback

Assessment information provide feedback both to teachers and students and guide them to take the further steps that are necessary to improve learning.

iii) Next Steps in Teaching and Learning'

Further steps to be taken next in teaching and learning are determined on the basis of the assessment information.

a) Guide students to know and execute the further measures that are needed to improve their learning.

b) Guide the teachers to introduce necessary modifications in their teaching according to the learning needs of students, identified by assessment information.

Summing up, assessment for learning helps students to know their present status of learning and plan the ways and means of improving it. As assessment of student learning, particularly formative assessment guides the students to improve their learning, it increases their confidence in learning.

4:11 Concept of Inclusive Practices

Before start teaching the disabled childrens, it is necessary to gather information from different sources, to assess extensively the nature and extent of their disabilities. Achievement tests, informal measures obtained from observation, pupil's self-report, report of the parents, information obtained from continuously monitoring the learning of children, report submitted by a team consisting of a doctor, special education teacher and the class teachers etc. form the sources of information to determine the nature and extent of disability level of learning and the efficiency of learning of disabled children.

Before implementing the contents of curriculum meant for a particular class/grade level, undertaking assessment of disabled children helps a great deal in knowing their learning needs and strengths and accordingly make available adequate support and special services [E.g. providing for special seats, enough space for moving in wheel chairs, Braille books and writing device, magnifying lenses, hearing aids, support staff etc.].

According to the nature and extent of disability of each child, **'Individual Education Plan'** (IEP) will have to be prepared by the class teacher, based on which in the inclusive classroom different instructional

techniques such as 'Group interaction sessions', 'Cooperative learning', 'Collaborative learning', making use of the services of the special education teacher in individualized instruction etc. will be used.

It is necessary to use appropriate methods and materials during and after teaching, to assess the learning achievement / performance expressed by disabled children. The achievement tests used for normal children can not be used for disabled children. For assessing the learning outcomes of each disabled child, appropriate assessment methods are to be employed using suitable tools and techniques, of which the following are the important ones.

i) Assessing the oral expressions of learning outcomes.

ii) Assessing the listening performance.

iii) Assessing the performance of the students through written expressions.

iv) Assessing the basic reading skills.

v) Assessing reading comprehension.

vi) Assessing the numerical ability and computational skill.

vii) Assessing the skills to solve problems using mathematical computations.

viii) Assessing the understanding of science concepts.

4:12 Assessing the Performance Outcomes of Diverse Learners

It is necessary to employ various methods and techniques of assessment of learning attainment / performance outcomes of different types of students like the gifted, slow learners, normal pupils, students having different kinds of physical disabilities, the mentally challenged, those having learning disabilities, those belonging to the disadvantaged and poor families etc. all learning together in the same class.

Let us discuss in detail the assessment of learning outcomes of different kinds of pupils, particularly different types of disabled pupils, all learning together with normal children in an inclusive classroom.

4:12:1 Assessing of the Learning Outcomes of the Visually Impaired

Those who are totally blind and those with significant loss of visual power that could not be corrected through the use of spectacles are called **'Visually Impaired'.**

Totally blind students learn by using their auditory and kinesthetic (touching) sensations. Students with moderate visual impairment could learn with the help of large print text books, magnifying glass and powerful table lamp.

4:12:1:01 Formative Assessment

The teacher can assess the learning level of students by asking oral questions in the content areas that are taught in the class and receive their reply. If home assignments are provided, questions could be written on a sheet of paper and given to the visually impaired students for which they could prepare the answers with the assistance of their parents or scribes and submit the same to the teacher. Further, for the submission of assignments, visually impaired students can be given additional time say a few days more than that given to normal students. The teacher can assess the participation of visually impaired students in classroom group discussions and use it in the formative assessment of performance outcomes.

4:12:1:02 Summative Assessment

Generally in summative examinations, the question papers given to visually impaired students are the same as those given to normal students. However, each visually impaired student will be seated in a separate place and the answers given out orally by him/her will be recorded verbatim by one of the scribes appointed for this purpose. Further, additional time will also be given for such students, to complete dictating their answers to the question paper given to them.

4:12:2 Auditorily Impaired

One who loses his/her hearing capacity after acquiring speech is known as **'hard of hearing'.** Those who are born with less capacity of hearing or those who lost their hearing capacity in the childhood before acquiring the ability to speak are known as **'deaf'**. If a child could hear sound of 50 decibels (decibel is the unit of measuring the intensity of sound and one decibel is the least possible intensity of sound that could be sensed by a human ear) and more only, then the child is auditorily handicapped, requiring special provisions for its education. In the education of auditorily handicapped children, oral education forms an important component. Lot of patience is required to impart instruction to auditorily handicapped children; it consumes lot of time also. In the teaching of auditorily handicapped children, visual and tactile stimuli are increasingly used. **'Lip reading'** and **'Sign Language'** are also taught to the auditorily handicapped children to participate in communication with others.

Children who are 'hard of hearing' if identified at an early age through the use of medical tests, their condition of hearing could be improved with the use of **'hearing-aids'**.

4:12:2:01 Assessing the Learning Outcomes of Auditorily Handicapped Children

Assessment of learning outcomes of auditorily handicapped children consists of two criteria. They are :

i) Communicating the criteria of assessment to the auditorily handicapped students through gestures, facial expression, using objects, pictures or printed sheets etc.

ii) Judging the level of achievement of each stage of the assessment task and note it.

Summative Assessment

For auditorily handicapped pupils, printed question papers can be used. To ensure that the student has understood each question, measures like facial expressions, gestures, lip reading, sign language etc. could be used to communicate with them. It should be kept in mind that the auditorily handicapped people generally use **'Visual Cues'** to understand the information communicated to them orally.

4:12:3 Learning Disabilities

'Learning disability' is a neuro-behavioural disorder affecting brain's ability to receive and process information. It is generally seen as a **Specific developmental disorder** affecting basic skills like reading, writing, numerical computations. etc. Children with such disability will not be able to participate in regular classroom activities as compared to other normal children. Learning disability in reading, writing and numerical computations are respectively called as '**dyslexia**', '**dysgraphia**', and '**discalculia**'. The disability to move one's bodily organs as he thinks is called '**dyspraxia**'. The disability due to not able to rotate the tongue properly is known as '**dysphasia**'. Generally, for those having learning disability, only formative assessment and no separate summative assessment is undertaken. Learning progress is recorded in the cumulative record.

Learning assessment of students with learning disabilities involves the following techniques :

i) Making each disabled child to function according to his/her level of skill development.

ii) Providing more drill and practice.

iii) After achieving learning proficiency in the basic skills required for a certain level of performance, moving to the next higher level.

iv) Using the carefully prepared individual instructional plan, increasing the level of every small learning skill.

v) Increasing the teacher-pupil interactions

vi) Correcting the student's mistakes in learning, then and there

vii) Grouping students according to the levels of learning attainment

viii) Assessing the learning progress often

ix) Using the computer that can analyse words, identify the proper spelling and contains a dictionary too.

x) Making use of the computer programmes which converts speech into the written form and vice-versa

xi) Making use of the 'talking calculators'

xii) Using books on audio-tape

xiii) Undertaking computer-based activities

xiv) Providing the facility for making available the services of note-taker, reader, proof-reader, scribe etc. needed for the teaching and assessing the learning of children with learning disabilities

xv) Making the students to spend a specific amount of time every day in the resource room

xvi) Allowing students to use 'addition and multiplication tables' during the assessment tests

xvii) Providing more space in the answer sheets for rough work in solving numerical problems

xviii) While evaluating student answers, without taking into consideration the spelling and grammatical mistakes committed, assessing only the content which they try to convey.

xix) In assessment tests, instead of giving questions requiring long answers, converting them into questions each of which require very small answers.

4:12:4 The Mentally Challenged

Those with I.Q. scores less than 70 in a standardised intelligence test are called 'mentally challenged'. They were previously known by the name 'Mentally retarded'. These children besides having sub-average general intellectual functioning, also lag behind significantly in their physical, emotional and social development as compared to children of same age. In teaching the 'Educably mentally retarded children', apart from giving them more time to do any task or answering questions, use of demonstrations, learning by doing along with the teacher, use of reinforcers like visual and tactile cues etc. are to be increasingly used. An 'Individual Educational Plan' (IEP) should be prepared and accordingly the programme of teaching and assessing should be implemented.

Giving training in choosing the correct answer among the three provided and matching it with the stimulus given, speaking, writing words and numbers correctly, painting with different colour options etc. and then assessing the performance of each child separately should be undertaken.

4:12:5 Other Categories of Students

For normal children as well as slow learners, orthopaedically handicapped and the gifted in the class, assessment of learning and achievement should be made through written tests besides conducting oral tests and practical tests. For those with orthopaedic handicap, appropriate provisions should be made available such as easy access to their classroom, therapeutic measures to move their limbs properly or remedial measures to minimize their forms of motor disabilities so that they could gain agility and functional efficiency.

Briefly stated, pupils with different kinds of disabilities such as the physically challenged, the mentally challenged, the orthopaedically handicapped, pupils with learning disabilities, slow learners etc. should also be provided with suitable opportunities and facilities that are available to normal pupils, to express their learning achievement and get them assessed.

4:13 Assessment and Feedback

Feedback refers to providing students with the knowledge of results obtained in the assessment test, based upon which they are informed of the deficiencies found in their learning, along with the recommendations for removing them. The primary purpose of assessment is to improve student learning. For this, the feedback followed by assessment helps a lot.

Generally, feedback is provided both after the formative assessments that are undertaken during the process of teaching and the summative assessments that are held after the teaching ended.

Following the feedback provided based on the formative assessment test, the teacher and student together plan the ways and means of eliminating the learning deficiencies pointed out. Students try to improve their learning by adopting measures like the modifications needed in their learning methods, undertaking additional drill and practice, discussing with the teacher, peers and their parents and try to utilize more learning resources etc. The teacher will also based on the feedback, try to incorporate necessary changes in his teaching method; use some additional instructional aids and devices. Stated briefly, feedback provided following the formative assessment

will be useful to both sides, the teacher and students. This kind of feedback results in the improvement of learning of all kinds of students. Self-confidence, self-esteem and learning motivation of all students will increase.

Following the summative assessment, students who lag behind in learning will be identified, based on their learning achievement and diagnostic test is conducted for them. Based on the findings of the diagnostic test, the teacher will undertake remedial measures and to make the students understand the concepts in the subject areas in which they have difficulties in learning.

All the remedial measures are not alike; they differ according to the nature and causes for the learning difficulties experienced by students. Some kind of learning difficulties could be removed by review and revision; some other learning difficulties may need **'remedial teaching'.** Measures like giving additional explanations and illustrations, making use of some more instructional materials and resources, providing individual attention to each student, increasing the participation of students in the teaching-learning process, increasing students' drill and practice etc. may find place in 'remedial teaching'.

For some learning difficulties adopting measures like increasing learning motivation in students, solving their emotional problems, controlling the external environmental factors that affect student learning etc. may be necessary.

Summing up, assessment of student learning and providing feedback are related to each other. **Without assessment, providing feedback is not possible; assessment without feedback is ineffective.**

4:14 Process of Feedback

Feedback serves as the connecting bridge between the assessment of student learning by the teacher and following that the measures he takes to improve students' learning.

The elements involved in the process of feedback viz.

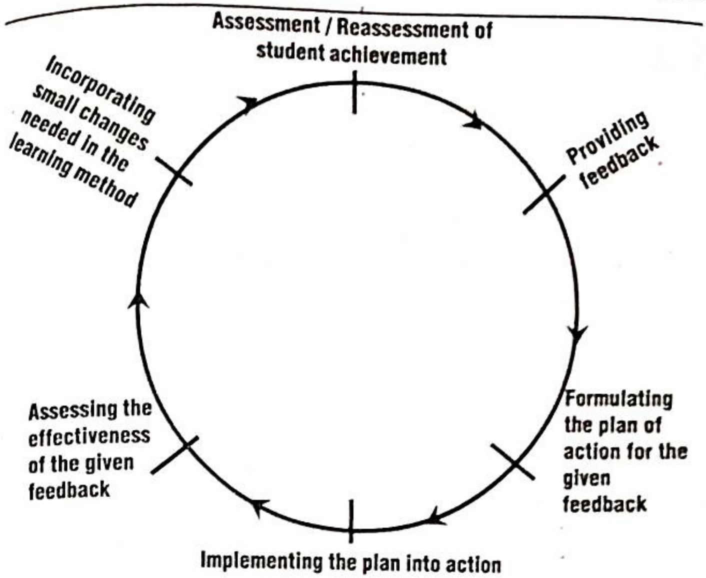
i) Formulating the plan of action for the given feedback

ii) Implementing the plan of action

iii) Assessing the effectiveness of the given feedback

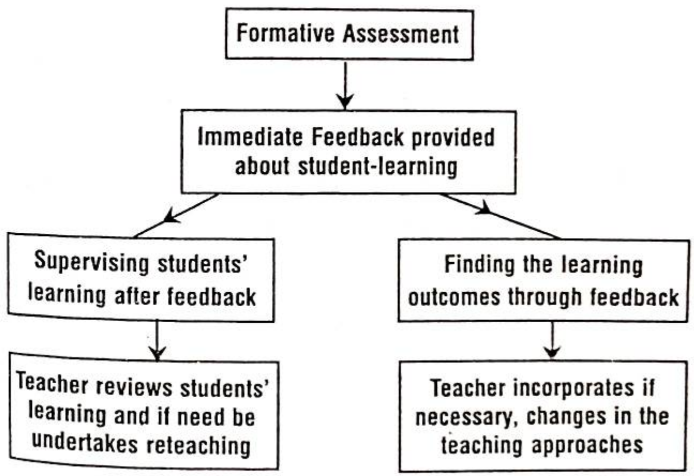
iv) Incorporating small changes needed in the learning method and

v) Reassessment of the learning achievement of students are sequential and related successively to form a cycle, as shown in the following diagram.



4:14:1 Feedback Process in Formative Assessment

In the formative assessment, providing feedback is integrated in the teaching-learning process as follows :



4:15 Feedback Strategies

The strategies that are used in the feedback given to students, following a formative evaluation test, so as to improve student learning, are discussed below :

4:15:1 Evaluative Strategies

Generally, this kind of strategies express approval or disapproval for students' performance / response for test-items, which may be in verbal, non-verbal or written forms. Verbal expressions of teacher's approval include use of such phrases like 'explained well', 'good performance', 'very good', 'excellent performance', 'simply brilliant' etc. Verbal expressions of teacher's disapproval include the use of phrases like 'wrong answer', 'irrelevant answer', 'you are writing by sheer guess', 'you are pouring out rubbish' etc. Non-verbal expressions of teacher's approval / recognition includes acts like 'teacher moving his head up and down', 'smiling', 'patting on the back', 'clapping hands', etc. Teacher's act like 'staring hard', 'pulling faces', 'moving the head laterally on both sides', etc. are expressions of his disapproval that may find place in feedback.

The expression of the teacher's approval / disapproval is a form of evaluative feedback strategy. Generally teachers offer their approval / recognition for correct answers, good performance, independent

thinking, creative thoughts etc. Teacher's disapproval is usually expressed for students' behaviour related response.

4:15:2 Descriptive Feedback Strategies

The following strategies are employed in providing feedback about student performance in the descriptive form.

i) Telling the Student / students his or their response is Correct or Wrong

This kind of feedback merely telling the student / students whether the answer advanced is right or wrong, lacks proactiveness. This kind of feedback is described as mechanical.

ii) Stating why an answer is Wrong

While assessing the answer scripts, when the teacher starts writing the feedback comments he can complement the students for their good answers by using descriptive phrases like 'well explained', 'extensive answer is given', 'having good vocabulary', 'good language style etc. Similarly for indicating the mistakes committed, the teacher should specifically state what type of mistakes have been done and how to avoid or rectify them. Such a kind of descriptive feedback alone will prove to be effective in improving student learning and performance.

iii) Stating what Students have and what they have not achieved

In this kind of feedback strategy, the teacher will make the student know how far he / she has been successful in his / her learning in relation to a specific goal, in what aspects of learning success have not been achieved and the reasons for the same.

iv) Specifying a better way of expressing a response

In this kind of feedback strategy, the focus is to explain to students the modifications required in improving their learning and how could they be accomplished.

v) Feedback suggesting ways to improve one's learning

In this kind of feedback strategy the focus is, after assessing the learning performance of each student, based on which the teacher points out the methods of improving learning and among those the one that suits well for a particular student.

Conclusion

In this unit, Meaning and definition of differential assessment, principles of differential assess-ment, Concept of culturally responsive assessment, ways

and means of organizing culturally responsive assessment, Use of tests for learner appraisal, Steps involved in the Construction of an achievement test and standardizing it, differences between a teacher made achievement test and a standardized achievement test, Construction of a diagnostic test, Scoring Key and Marking Scheme, Qualities of a good test, Item analysis, Ensuring fairness in assessment, Assessment for enhancing students' confidence in learning, assessing the disabled and performance outcomes of diverse learners (Visually impaired, auditorily impaired, those having different kinds of learning disabilities, the mentally challenged, orthopeadically handicapped etc.), assessment and feedback, process of feedback, and feedback strategies have been discussed.

Review Questions:

1. What is meant by 'Differential assessment'? (C) [Ans: 4:01 + 4:01:1]
2. What do you mean by differential assessment? Explain the principles of differential assessment. (B)[Ans: 4:01:1 + 4:01:2]

3. Explaining the concept of culturally responsive assessment, describe the ways and means of organizing it. (A) [Ans: 4:02 + 4:02:1]

4. Explain how tests could be used in learner appraisal and their important features. (B) [Ans: 4:03 + 4:03:1]

5. Discuss the steps involved in the construction of an achievement test. (A) [Ans: 4:04 + 4:04:1 to the end of 4:04:4]

6. Explain with an illustration the method of constructing a diagnostic test. (A) [Ans: 4:06 + 4:06:1 + 4:06:1:01 to the end of 4:06:1:04 + 4:06:2]

7. Describe the method of diagnosing the nature of learning difficulties of students and taking remedial measures. (A) [Ans: 4:06:4:1 + 4:06:5]

8. Describe the method of undertaking item analysis of a question paper and mention its uses. (B) [Ans: 4:07 + 4:07:1]

9. Explain the qualities of a good test. (B) [Ans: 4:08 + 4:08:1 to the end of 4:08:6]

10. Explain the concept of 'ensuring fairness in assessment' and the steps to be taken to make assessment practices fair to all students. (A) [Ans : 4:09:1 + 4:09:2]

11. Explain the assessment for enhancing students' confidence in learning. (A) [Ans: 4:10]

12. Explain the common methods of assessing the learning of disabled children. (B) [Ans: 4:11]

13. Describe the methods of assessing the learning performance of diverse learners. (A) [Ans: 4:12 + 4:12:1 + 4:12:1:01 + 4:12:1:02 + 4:12:2 + + 4:12:2:01 + 4:12:3 to the end of 4:12:5]

14. Explain how to ensure assessment of learning and feedback to students in an inclusive classroom are integrated to each other. (B) [Ans: 4:13]

15. Explain the process of 'providing feedback to students'. (A) [Ans: 4:14 + 4:14:1 + 4:15 + 4:15:1 + 4:15:2]